



FOSSEE Semester Long Internship - Autumn 2026

Task 4 — eSim Upgradation (CSE and Related Fields)

Screening Task Report

Submitted by

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System / Project Details

Field	Value
Candidate	Neeraj Kumar
Operating System	Ubuntu 25.04 (Plucky), 64-bit
eSim Version	2.5
Repository Branch	installers
Fork	github.com/NeerajKumar011/eSim

1. Introduction

This report documents the installation and debugging process of eSim 2.5 on Ubuntu 25.04 as part of the FOSSEE eSim Task 4 screening task.

The objective of this task was to:

- Install eSim 2.5 on Ubuntu 25.04
- Identify the installation and compatibility issues encountered during the process
- Investigate the root cause of each issue
- Modify the installer scripts to resolve the identified problems
- Verify each correction and continue the installation successfully
- Document the complete troubleshooting process in a clear and reproducible manner

During the installation process, six major issues were encountered in sequence:

- Missing fi in the main Ubuntu installer script
- Incorrect configuration commands in install-eSim.sh, causing /: Is a directory
- Premature termination when KiCad 8.0 was already installed
- NGHDL rejecting Ubuntu 25.04 as an unsupported version
- Unavailable libcanberra-gtk-module package on Ubuntu 25.04
- GHDL 4.1.0 rejecting the LLVM 20.1.2 version available on Ubuntu 25.04

Each issue was investigated individually and addressed with a targeted modification to the installation workflow. After applying the corrections, the installation completed successfully, and the installed eSim, GHDL, Ngspice, Verilator, and KiCad components were subsequently verified.

2. Environment and Repository Setup

Item	Observed value
OS	Ubuntu 25.04 (Plucky), 64-bit
Kernel	6.14.0-37-generic
Python	3.13.3
Git	2.48.1
eSim	2.5

KiCad	8.0.8
Ngspice	35
GHDL	4.1.0
LLVM	20.1.2
Verilator	4.210

2.1 Repository preparation

The work was carried out on the candidate's fork of eSim. The repository was ultimately used on the installers branch, and the final changes were pushed over SSH.

Commands used:

```
git clone git@github.com:NeerajKumar011/eSim.git
cd eSim
git checkout installers
```

Verify repository and branch:

```
git remote -v:
origin  git@github.com:NeerajKumar011/eSim.git (fetch)
origin  git@github.com:NeerajKumar011/eSim.git (push)
git branch --show-current
installers
```

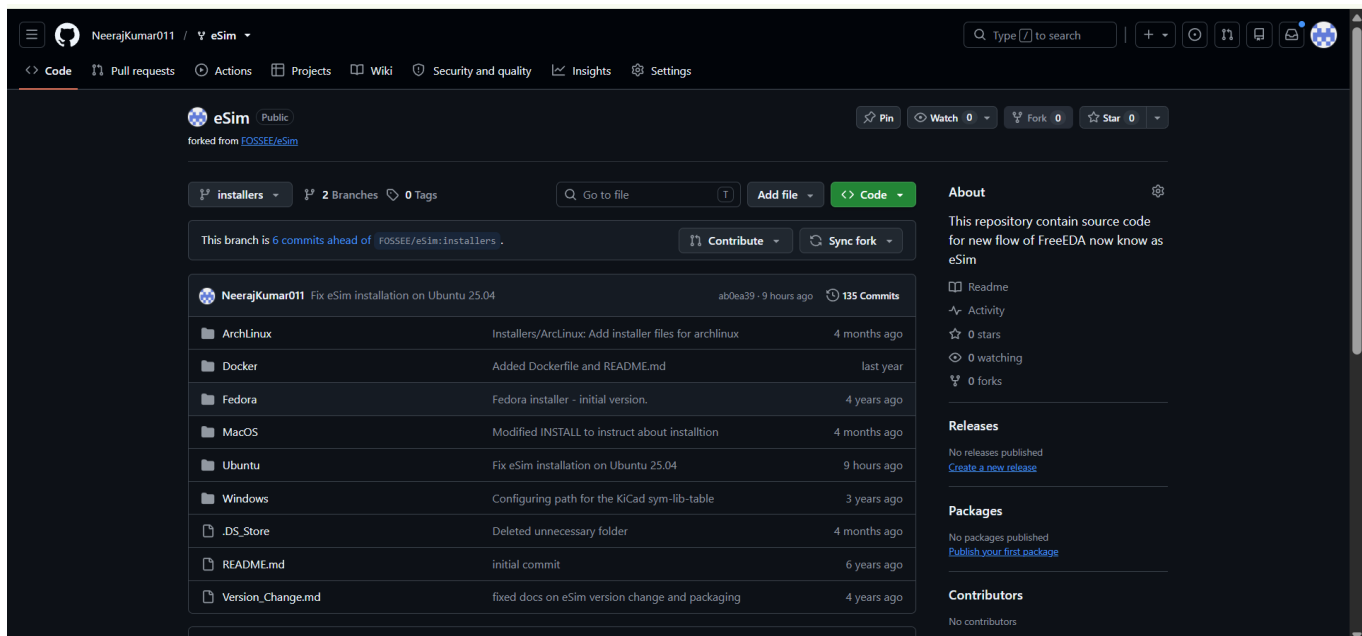


Figure 1. Candidate's eSim fork used for the submission work.

Commands used:

```
git clone git@github.com:NeerajKumar011/eSim.git
cd eSim
git checkout installers
```

Repository-based installation:

```
cd ~/eSim/Ubuntu
chmod +x install-eSim.sh
./install-eSim.sh -install
```

Released eSim 2.5 package:

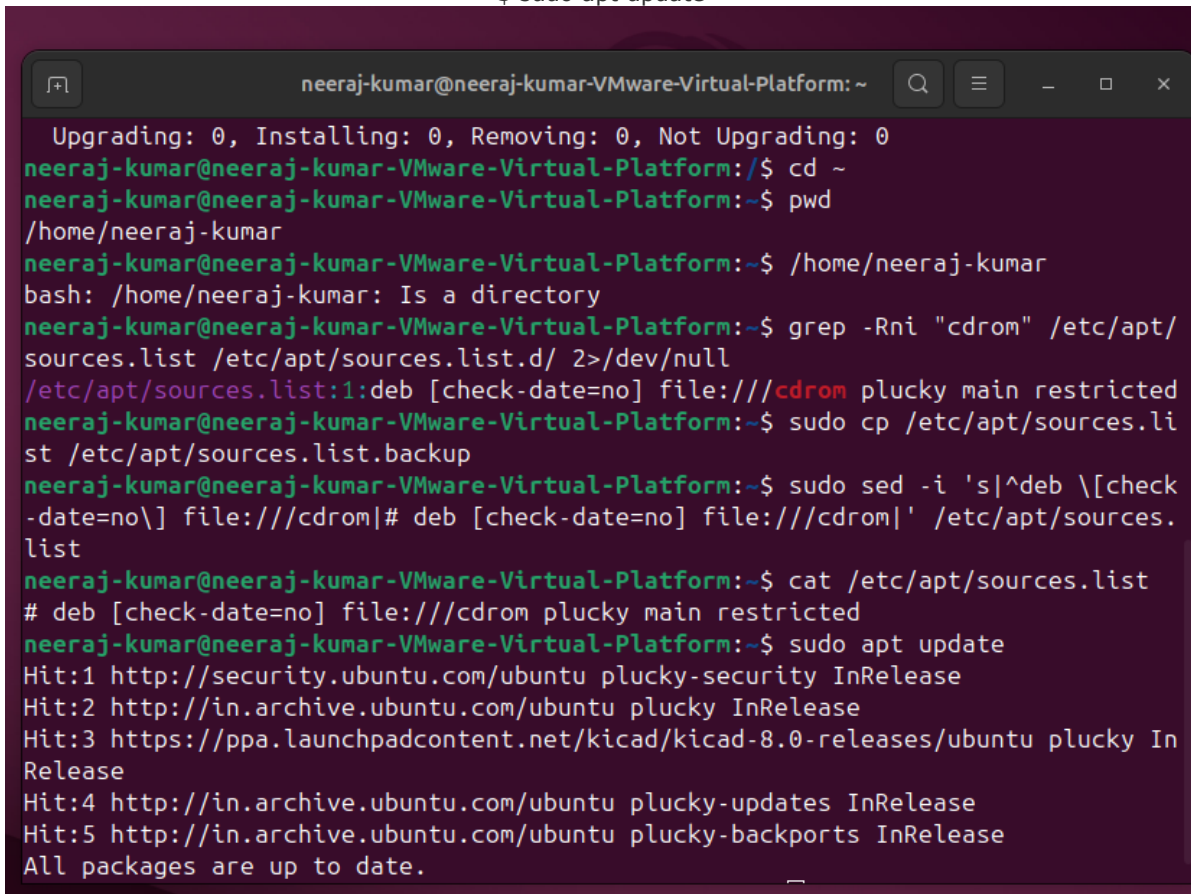
```
cd ~/Downloads/eSim-2.5
./install-eSim.sh --install
```

2.2 Ubuntu package source preparation

The VM initially contained a stale CD-ROM APT source. It was removed/commented so that normal package installation could proceed.

```
$ grep -Rni "cdrom" /etc/apt/sources.list /etc/apt/sources.list.d/ 2>/dev/null
$ sudo cp /etc/apt/sources.list /etc/apt/sources.list.backup
$ sudo sed -i 's|^deb .*file:///cdrom|# &|' /etc/apt/sources.list
$ sudo apt update
```

\$ sudo apt update



```
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform: ~
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:/$ cd ~
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~$ pwd
/home/neeraj-kumar
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~$ /home/neeraj-kumar
bash: /home/neeraj-kumar: Is a directory
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~$ grep -Rni "cdrom" /etc/apt/
sources.list /etc/apt/sources.list.d/ 2>/dev/null
/etc/apt/sources.list:1:deb [check-date=no] file:///cdrom plucky main restricted
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~$ sudo cp /etc/apt/sources.li
st /etc/apt/sources.list.backup
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~$ sudo sed -i 's|^deb \[check
-date=no\] file:///cdrom|# deb [check-date=no] file:///cdrom|' /etc/apt/sources.
list
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~$ cat /etc/apt/sources.list
# deb [check-date=no] file:///cdrom plucky main restricted
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu plucky-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu plucky InRelease
Hit:3 https://ppa.launchpadcontent.net/kicad/kicad-8.0-releases/ubuntu plucky In
Release
Hit:4 http://in.archive.ubuntu.com/ubuntu plucky-updates InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu plucky-backports InRelease
All packages are up to date.
```

Figure 2. APT update after disabling the stale CD-ROM repository.

3. Error-by-Error Method

Each installer error was handled in the same order:

1. Capture the exact error.
2. Identify the installer file/function responsible.
3. Apply only the smallest correction required for that error.
4. Run the syntax/command check needed to validate the correction.
5. Rerun the installer and confirm that this error is gone before handling the next error.

Error 1 — Missing `fi` in install-eSim.sh

Observed error:

```
$ ./install-eSim.sh: line 70: syntax error near unexpected token `}'
$ ./install-eSim.sh: line 70: `}'
```

```

neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/eSim/Ubuntu$ ./install-eSim.sh
sh --install
./install-eSim.sh: line 70: syntax error near unexpected token `}'
./install-eSim.sh: line 70: `}'
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/eSim/Ubuntu$ bash install-eS
im.sh --install
install-eSim.sh: line 70: syntax error near unexpected token `}'
install-eSim.sh: line 70: `}'
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/eSim/Ubuntu$ bash install-eS

```

Figure 3. Actual Error 1: Bash reports the unexpected `}`.

Why it happened

The top-level installer opened an `if [[ -f "\$SCRIPT" ]] block but did not close it with `fi` . Bash therefore reached the function's closing `}` while still inside an incomplete conditional.`

Correction steps

Run the following commands from the eSim Ubuntu installer directory, then rerun the installer.

```
$ sed -i '/bash "$SCRIPT" "$ARGUMENT"/a\ fi' install-eSim.sh
```

Verification

```
$ bash -n install-eSim.sh
$ echo $?

```

The syntax check returned 0, so Bash accepted the script structure. The corrected script was then used for the next installer run.

Note The actual source edit is one inserted line: ``fi`` immediately after the command that runs the selected installer script.

Error 2 — Misplaced configuration writes cause ``/: Is a directory``

Observed error:

```
$ ./install-eSim.sh: line 62: /: Is a directory
$ ./install-eSim.sh: line 63: /: Is a directory
$ ...
$ ./install-eSim.sh: line 68: /: Is a directory

```

```

neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/eSim/Ubuntu$ sed -n '55,80p'
install-eSim.sh
        ;;
    esac

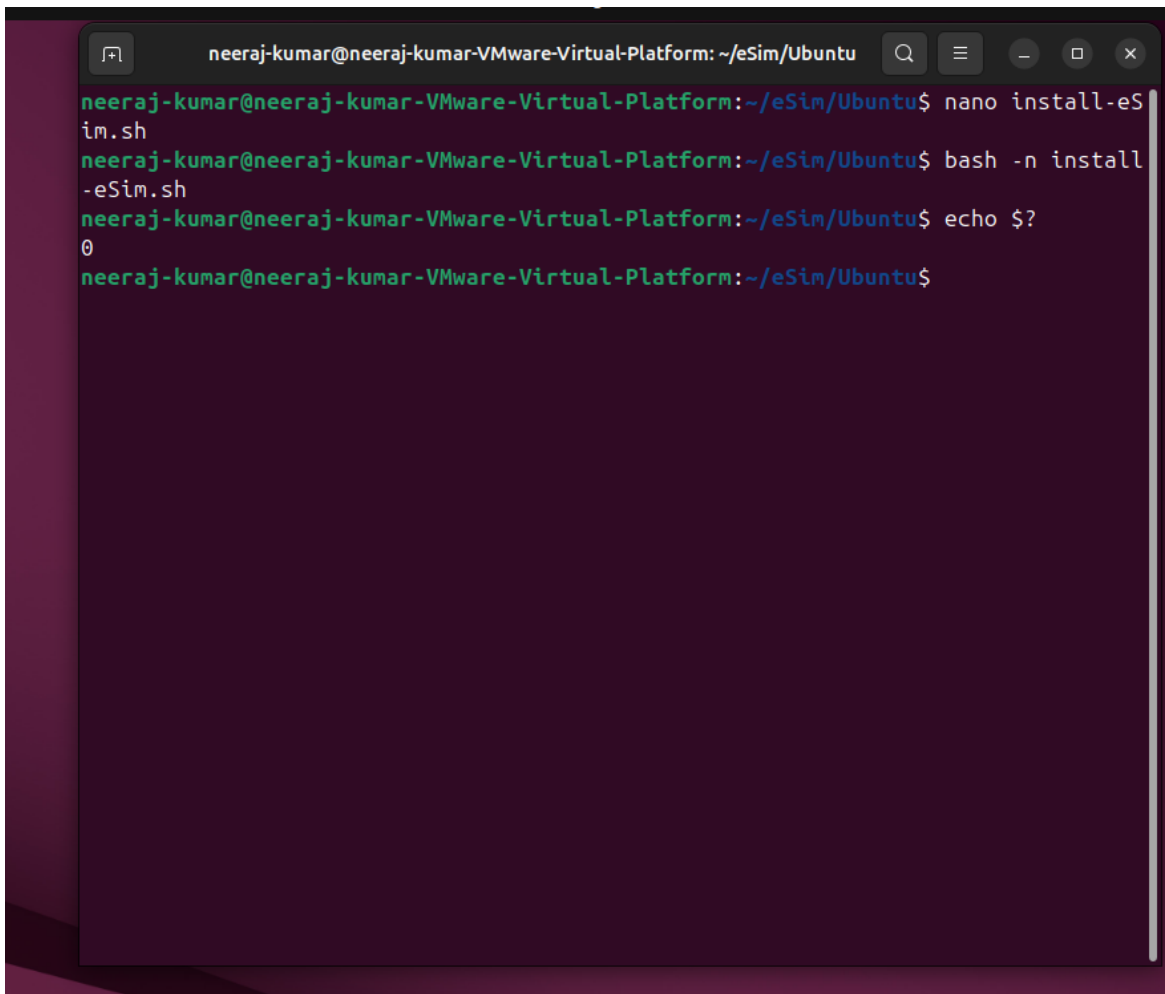
    # Run the script if found
    if [[ -f "$SCRIPT" ]]; then
        echo "Running script: $SCRIPT $ARGUMENT"
        bash "$SCRIPT" "$ARGUMENT"
        echo "[eSim]" >> $config_dir/$config_file
        echo "eSim_HOME = $eSim_Home" >> $config_dir/$config_file
        echo "LICENSE = %(eSim_HOME)s/LICENSE" >> $config_dir/$config_file
        echo "KicadLib = %(eSim_HOME)s/library/kicadLibrary.tar.xz" >> $config_dir/$
config_file
        echo "IMAGES = %(eSim_HOME)s/images" >> $config_dir/$config_file
        echo "VERSION = %(eSim_HOME)s/VERSION" >> $config_dir/$config_file
        echo "MODELICA_MAP_JSON = %(eSim_HOME)s/library/ngspicetoModelica/Mapping.js
on" >> $config_dir/$config_file
    }
}

function installNghdl
{
    echo "Installing NGHDL....."
    unzip -o nghdl.zip
    cd nghdl/
    chmod +x install-nghdl.sh
}

neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/eSim/Ubuntu$

```

Figure 4. The seven misplaced configuration writes visible in the top-level installer.



```
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform: ~/eSim/Ubuntu
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/eSim/Ubuntu$ nano install-eSim.sh
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/eSim/Ubuntu$ bash -n install-eSim.sh
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/eSim/Ubuntu$ echo $?
0
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/eSim/Ubuntu$
```

Figure 5. Actual Error 2: repeated `/: Is a directory` messages during installation.

Why it happened

The top-level `install-eSim.sh` contained seven configuration-writing commands that belonged in the Ubuntu-specific installer. Because the required configuration variables were not initialized there, their redirection target resolved incorrectly and Bash attempted to write to `/`.

Correction steps

Run the following commands from the eSim Ubuntu installer directory, then rerun the installer.

```
$ cd ~/eSim/Ubuntu
$ cp install-eSim.sh install-eSim.sh.before-config-fix
$ nano install-eSim.sh
```


Inside install-eSim.sh, the following seven lines were deleted manually:

```
echo "[eSim]" >> $config_dir/$config_file
echo "eSim_HOME = $eSim_Home" >> $config_dir/$config_file
echo "LICENSE = %(eSim_HOME)s/LICENSE" >> $config_dir/$config_file
echo "KicadLib = %(eSim_HOME)s/library/kicadLibrary.tar.xz" >> $config_dir/$config_file
echo "IMAGES = %(eSim_HOME)s/images" >> $config_dir/$config_file
echo "VERSION = %(eSim_HOME)s/VERSION" >> $config_dir/$config_file
echo "MODELICA_MAP_JSON = %(eSim_HOME)s/library/ngspicetoModelica/Mapping.json"
```

Verification of the code change:

```
$ bash -n install-eSim.sh
$ git diff --check
```

Verification of the actual fix:

```
$ cd ~/Downloads/eSim-2.5
$ ./install-eSim.sh --install 2>&1 | tee esim-install.log
```

The important result was that the previous:

```
./install-eSim.sh: line 63: /: Is a directory
...
./install-eSim.sh: line 69: /: Is a directory
```

messages no longer appeared, and the installer progressed to the next error, which was the KiCad exit 0 problem.

The syntax and whitespace checks passed. A fresh eSim 2.5 installer run no longer produced the `/: Is a directory` messages and proceeded to the KiCad stage.

Note This fix removes only the misplaced block. The configuration-writing logic remains in the Ubuntu-specific installer where the required variables are defined.

Error 3 — Installer exits when KiCad 8 is already installed

Observed error

```
Installing KiCad.....
Ubuntu 25.04 detected.
$ KiCad 8.0 is already installed.
```

The installer stopped immediately after this message. Source inspection showed that the `installKicad()` function used `exit 0` in the already-installed case.

Why it happened

The installKicad() function contained:

```
135 echo "Kicad 8.0 is already installed."  
136 exit 0  
137 fi
```

Because installKicad() is a function, exit 0 terminates the entire installer, instead of returning from that function. As a result, the following installation steps were never reached:

```
copyKicadLibrary  
installNghdl  
installSky130Pdk  
createDesktopStartScript
```

Correction steps

```
$ cd ~/eSim/Ubuntu  
$ sed -i '136s/exit 0/return 0/' install-eSim-scripts/install-eSim-25.04.sh
```

Verification

```
$ cd ~/Downloads/eSim-2.5  
$ bash -n install-eSim-scripts/install-eSim-25.04.sh  
$ ./install-eSim.sh --install 2>&1 | tee esim-install.log
```

Confirmation

After the correction, the installer no longer stopped after:

```
KiCad 8.0 is already installed.
```

It continued to:

```
kicad config folder already exists  
symbol table copied in the directory  
Installing NGHDL.....
```

The next installation run no longer stopped at the KiCad message and continued into the custom-library step and then NGHDL.

Source change The effective change is exactly `exit 0` → `return 0` in the already-installed KiCad branch.

Error 4 — NGHDL does not support Ubuntu 25.04

Observed error:

```
$ Detected Ubuntu Version:  
$ Unsupported Ubuntu version: 25.04 ()
```

After Error 3 was fixed, the main installer successfully reached NGHDL. The bundled NGHDL dispatcher only had explicit Ubuntu installer paths for older releases.

Why it happened

The NGHDL dispatcher checked specific Ubuntu releases and rejected unsupported versions. Ubuntu 25.04 therefore reached the unsupported-version branch.

Correction steps

The Ubuntu 25.04 eSim wrapper uses the bundled Ubuntu 24.04 NGHDL installer directly, instead of the NGHDL version dispatcher that rejects 25.04.

```
$ nano install-eSim-scripts/install-eSim-25.04.sh

Replace:
    ./install-nghdl.sh --install

With:
    bash install-nghdl-scripts/install-nghdl-24.04.sh --install
```

Verification

```
$ bash -n install-eSim-scripts/install-eSim-25.04.sh

$ git diff -- install-eSim-scripts/install-eSim-25.04.sh
```

The diff showed:

```
-    ./install-nghdl.sh --install
+    bash install-nghdl-scripts/install-nghdl-24.04.sh --install
```

and the Bash syntax check passed with no output.

Then the corrected installer was run against the actual eSim 2.5 package, and the installation proceeded into the NGHDL dependency installation instead of stopping at the Ubuntu 25.04 unsupported-version message. This confirmed Error 4 was fixed.

Error 5 — Unavailable GTK2 Canberra package

Observed error:

```
Installing Gtk Canberra modules.....
Error: Package 'libcanberra-gtk-module' has no installation candidate

Error! Kindly resolve above error(s) and try again.
Aborting Installation...
```

```

Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1
Installing Zlib1g-dev.....
zlib1g-dev is already the newest version (1:1.3.dfsg+really1.3.1-1ubuntu1).
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1
Installing Gtk Canberra modules.....
Package libcanberra-gtk-module is not available, but is referred to by another p
ackage.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

Error: Package 'libcanberra-gtk-module' has no installation candidate

Error! Kindly resolve above error(s) and try again.

Aborting Installation...

neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/nghdl$

```

Figure 7. Actual Error 5: Ubuntu 25.04 has no installation candidate for the legacy GTK2 Canberra module.

Why it happened

The bundled NGHDL Ubuntu 24.04 installer requested both `libcanberra-gtk-module` and `libcanberra-gtk3-module`. Ubuntu 25.04 provided the GTK3 package but no installation candidate for the legacy GTK2 package.

```
$ sudo apt install -y libcanberra-gtk-module libcanberra-gtk3-module
```

Ubuntu 25.04 does not provide the legacy libcanberra-gtk-module package, so APT could not satisfy the dependency.

Investigation

After encountering this error, the separate **FOSSEE/NGHDL repository** was downloaded so that the NGHDL installer could be inspected and the outdated dependency identified.

```

$ git clone https://github.com/FOSSEE/nghdl.git

$ cd ~/nghdl

$ git checkout installers

```

The eSim README also identifies NGHDL as the mixed-signal module and notes that the Ubuntu release package contains nghdl.zip.

Correction steps

Run the following commands from the eSim Ubuntu installer directory, then rerun the installer.

```
$ cd ~/eSim/Ubuntu
$ sed -i "s/libcanberra-gtk-module //" install-eSim-scripts/install-eSim-25.04.sh
```

This leaves **libcanberra-gtk3-module** in the dependency list.

Verification

First check the modified eSim installer:

```
$ bash -n install-eSim-scripts/install-eSim-25.04.sh
$ git diff --check
```

Then rerun the official eSim 2.5 package:

```
$ cd ~/Downloads/eSim-2.5

$ ./install-eSim.sh --install 2>&1 | tee esim-install.log
```

The dependency failure disappeared on the next installation run, and the process advanced to extracting/configuring GHDL. The next distinct failure was GHDL's LLVM 20.1.2 version check.

Note Only the obsolete GTK2 package reference is removed; the available GTK3 module remains requested.

Error 6 — GHDL 4.1.0 rejects LLVM 20.1.2

Observed error

```
Use full IEEE library
Build machine is: x86_64-linux-gnu
Unhandled version llvm 20.1.2

Error! Kindly resolve above error(s) and try again.
Aborting Installation...
```

```
ghdl-4.1.0 successfully extracted
Changing directory to ghdl-4.1.0 installation
Configuring ghdl-4.1.0 build as per requirements
gcc (Ubuntu 14.2.0-19ubuntu2) 14.2.0
Copyright (C) 2024 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Use full IEEE library
Build machine is: x86_64-linux-gnu
Unhandled version llvm 20.1.2

Error! Kindly resolve above error(s) and try again.

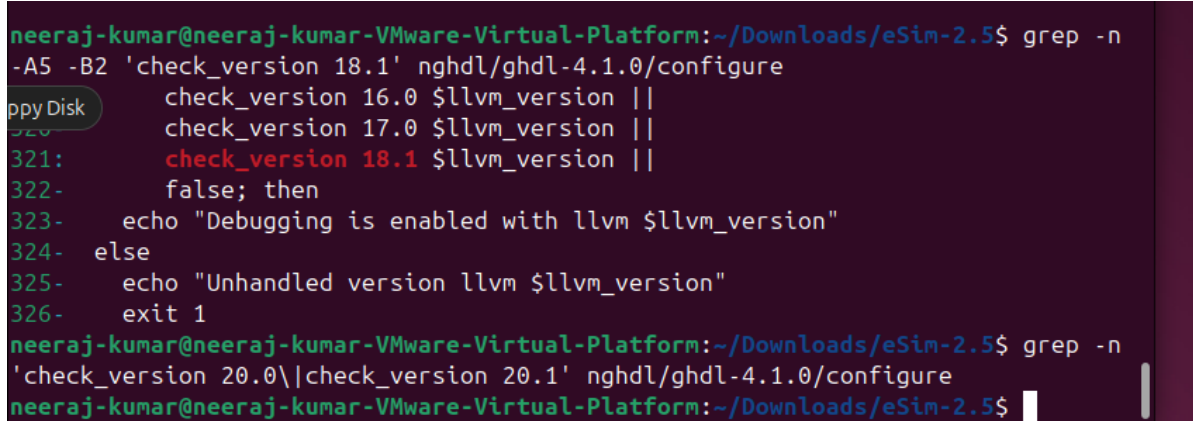
Aborting Installation...

neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/Downloads/eSim-2.5$
```

Figure 9. Actual Error 6: GHDL configure rejects LLVM 20.1.2.

Why it happened

The bundled GHDL 4.1.0 configure script checked LLVM versions only through 18.1 before falling through to the 'Unhandled version llvm ...' branch.

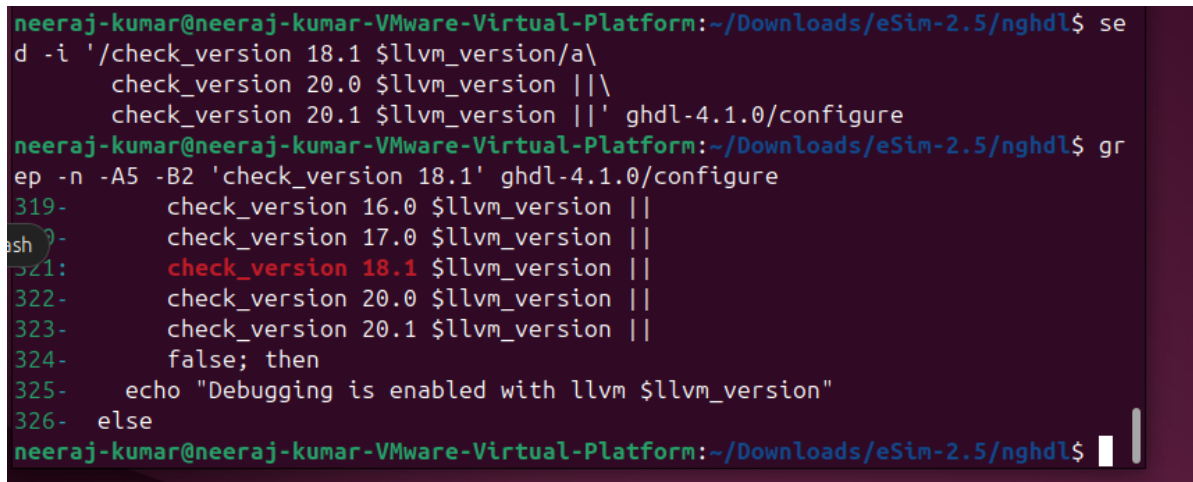


```
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/Downloads/eSim-2.5$ grep -n
-A5 -B2 'check_version 18.1' nghdl/ghdl-4.1.0/configure
320-     check_version 16.0 $llvm_version ||
321-     check_version 17.0 $llvm_version ||
322-     check_version 18.1 $llvm_version ||
323-     false; then
324-     echo "Debugging is enabled with llvm $llvm_version"
325- else
326-     echo "Unhandled version llvm $llvm_version"
327-     exit 1
328-
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/Downloads/eSim-2.5$ grep -n
'check_version 20.0|check_version 20.1' nghdl/ghdl-4.1.0/configure
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/Downloads/eSim-2.5$
```

Figure 10. Original GHDL configure version checks stop at LLVM 18.1.

Correction steps

```
$ cd ~/eSim/Ubuntu
$ sed -i '/chmod +x configure/a\
    sed -i '\''s/check_version 18.1 $llvm_version ||/&\
        check_version 20.0 $llvm_version ||\
        check_version 20.1 $llvm_version ||/'\'' configure' install-eSim-scripts/install-eSim-
25.04.sh
```



```
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/Downloads/eSim-2.5/nghdl$ se
d -i '/check_version 18.1 $llvm_version/a\
    check_version 20.0 $llvm_version ||\
    check_version 20.1 $llvm_version ||' ghdl-4.1.0/configure
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/Downloads/eSim-2.5/nghdl$ gr
ep -n -A5 -B2 'check_version 18.1' ghdl-4.1.0/configure
319-     check_version 16.0 $llvm_version ||
320-     check_version 17.0 $llvm_version ||
321-     check_version 18.1 $llvm_version ||
322-     check_version 20.0 $llvm_version ||
323-     check_version 20.1 $llvm_version ||
324-     false; then
325-     echo "Debugging is enabled with llvm $llvm_version"
326- else
327-     echo "Unhandled version llvm $llvm_version"
328-     exit 1
329-
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:~/Downloads/eSim-2.5/nghdl$
```

Figure 11. The inserted LLVM 20.0 and 20.1 checks are confirmed in the extracted GHDL configure file.

Verification

```
$ bash -n install-eSim-scripts/install-eSim-25.04.sh
$ git diff --check
$ which ghdl
$ ghdl --version

$ /usr/local/bin/ghdl
GHDL 4.1.0 (tarball) [Dunoon edition]
```

```
$ Compiled with GNAT Version: 14.2.0
$ llvm 20.1.2 code generator
```

A minimal VHDL design was then analyzed and elaborated with the installed compiler, and the command returned exit code 0.

```
ghdl -a /tmp/test.vhd
ghdl -e test
echo $?
0
```

Why this is the main technical fix The Ubuntu compiler stack was not downgraded. The installer was adapted to the LLVM version actually provided by Ubuntu 25.04, and the resulting GHDL installation was functionally tested.

5. Final Installation Result

After the six installer corrections, the official eSim 2.5 installation reached the final success message.

```
$ NGHDL installed successfully
$ ...
$ ----- eSim Installed Successfully -----
$ Type "esim" in Terminal to launch it
```

6. Final Functional Verification

6.1 Installed executables

```
which esim
/usr/bin/esim

which ngspice
/usr/bin/ngspice

which ghdl
/usr/local/bin/ghdl

which verilator
/usr/local/bin/verilator
```

6.2 GHDL

```
ghdl --version
GHDL 4.1.0 (tarball) [Dunoon edition]
llvm 20.1.2 code generator

ghdl -a test.vhd
ghdl -e test
# exit code: 0
```

6.3 Verilator

```
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:/tmp$ cd /tmp

cat > test.v <<'EOF'
module test;
endmodule
EOF

verilator --lint-only test.v

echo "Verilator exit code: $?"
Verilator exit code: 0
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:/tmp$
```

Figure 12. Verilator lint test completed with exit code 0.

6.4 Ngspice

An empty `ngspice -b` invocation returned exit code 1 because no circuit was supplied. That is an incomplete test input, not an installation failure.

A real resistor-divider circuit was then used:

```
V1 in 0 5
R1 in out 1k
R2 out 0 1k

.op
.print op v(in) v(out)

.end
```



```

Resistor models (Simple linear resistor)
model      R
rsh        0
narrow     0
short      0
tc1        0
tc2        0
tce        0
defw       1e-05
l          1e-05
kf         0
af         0
r          0
bv_max     1e+99
lf         1
wf         1
ef         1

Resistor: Simple linear resistor
device     r2          r1
model      R          R
resistance 1000        1000
ac         1000        1000
dtemp      0          0
bv_max     1e+99       1e+99
noisy      1          1
i          0.0025      0.0025
p          0.00625     0.00625

Vsource: Independent voltage source
device     v1
dc         5
acmag      0
pulse      -
sin        -
exp        -
pwl        -
sffn       -
am         -
trnoise    -
trrandom   -
i          -0.0025
p          -0.0125

* simple dc test
Operating Point Sat Aug 22 11:19:44 2026
-----
Index  v(in)      v(out)
-----
0      5.000000e+00 2.500000e+00

```

Figure 13. Real Ngspice resistor-divider simulation: $v(\text{in})=5\text{ V}$, $v(\text{out})=2.5\text{ V}$, exit code 0.

6.5 KiCad

```

=== NGSPICE TEST ===
Note: No ".plot", ".print", or ".fourier" lines; no simulations run

exit=1

=== GHDL TEST ===
exit=0

=== VERILATOR TEST ===
exit=0

=== KICAD CLI ===
/usr/bin/kicad-cli
8.0.8
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:/tmp$
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:/tmp$

```

Figure 14. Combined final verification showing GHDL and Verilator exit code 0 and KiCad CLI 8.0.8.

6.6 eSim GUI and project creation

```
p -0.0125
* simple dc test
Operating Point Sat Aug 22 11:19:44 2026
-----
index v(in) v(out)
-----
5.000000e+00 2.500000e+00

total analysis time (seconds) = 0.000404
total CPU time (seconds) = 0.021
total DRAM available = 3354.480 MB.
RAM currently available = 163.418 MB.
maximum ngspice program size = 17.477 MB.
current ngspice program size = 10.211 MB.

shared ngspice pages = 8.527 MB.
text (code) pages = 4.938 MB.
stack = 0 bytes.
library pages = 1.879 MB.

ngspice exit=0
neeraj-kumar@neeraj-kumar-VMware-Virtual-Platform:/tmp$
```

Figure 15. eSim workspace selection dialog.

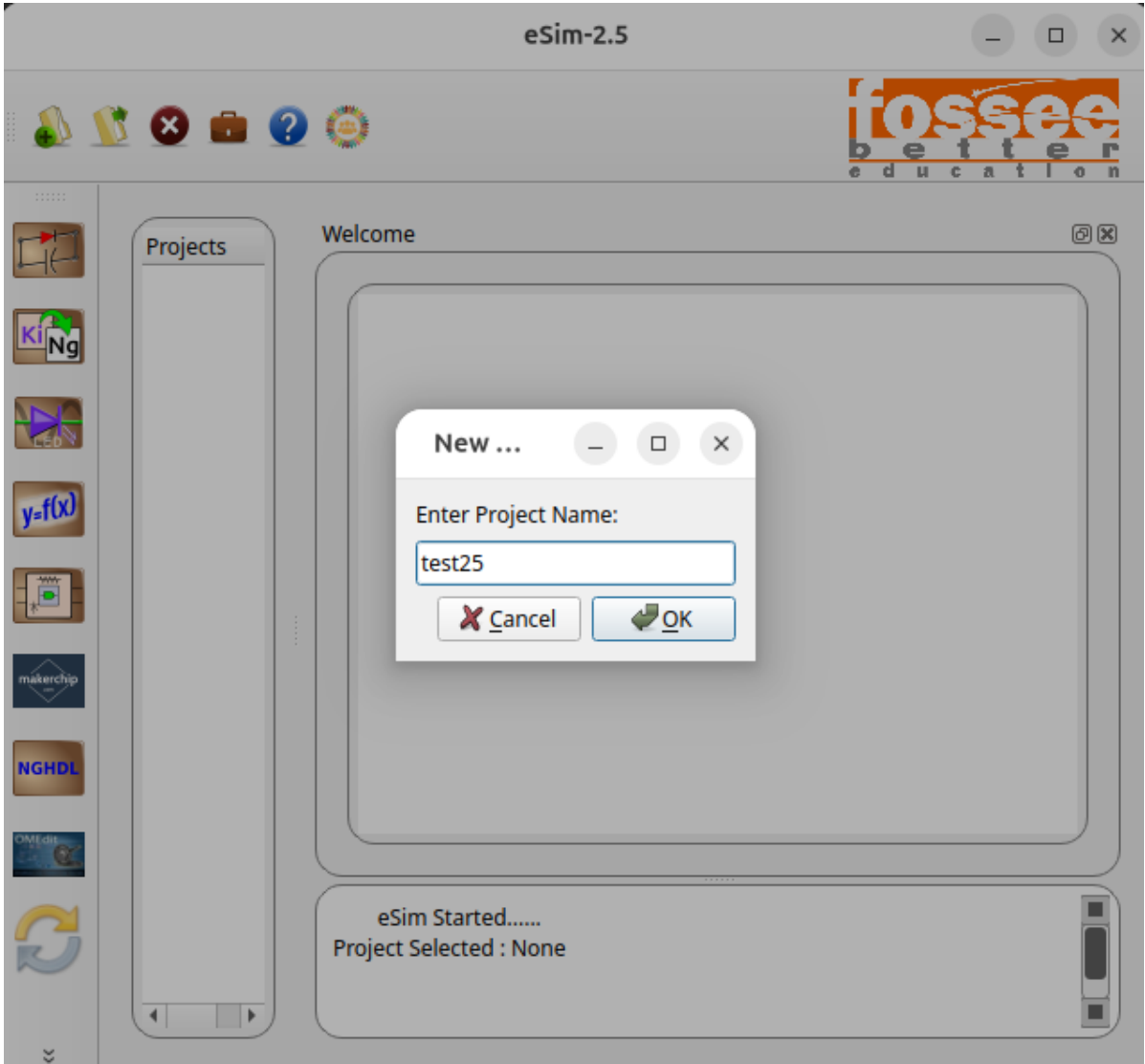


Figure 16. eSim project `test25` created successfully.

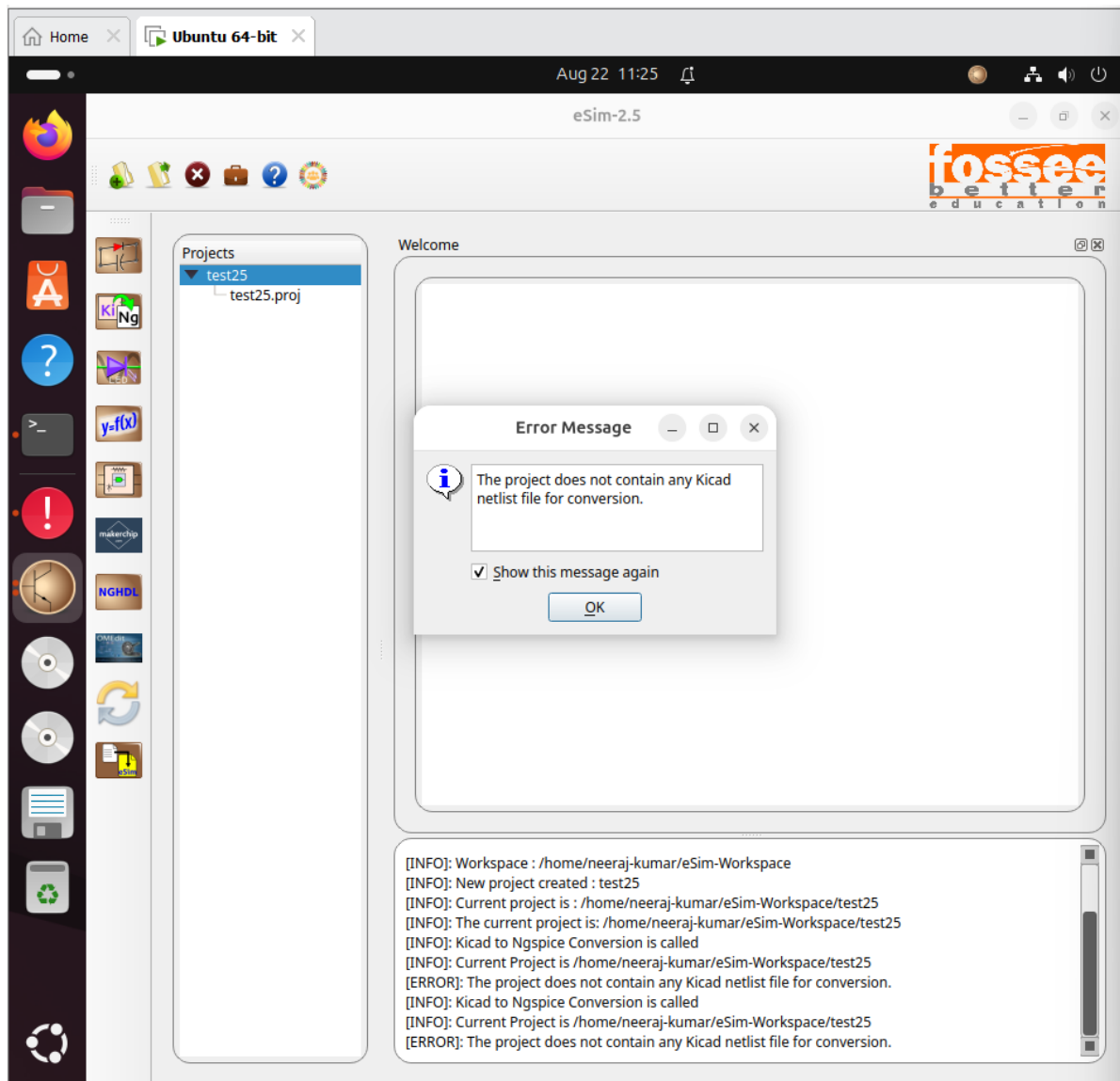


Figure 17. KiCad first-run symbol library configuration.

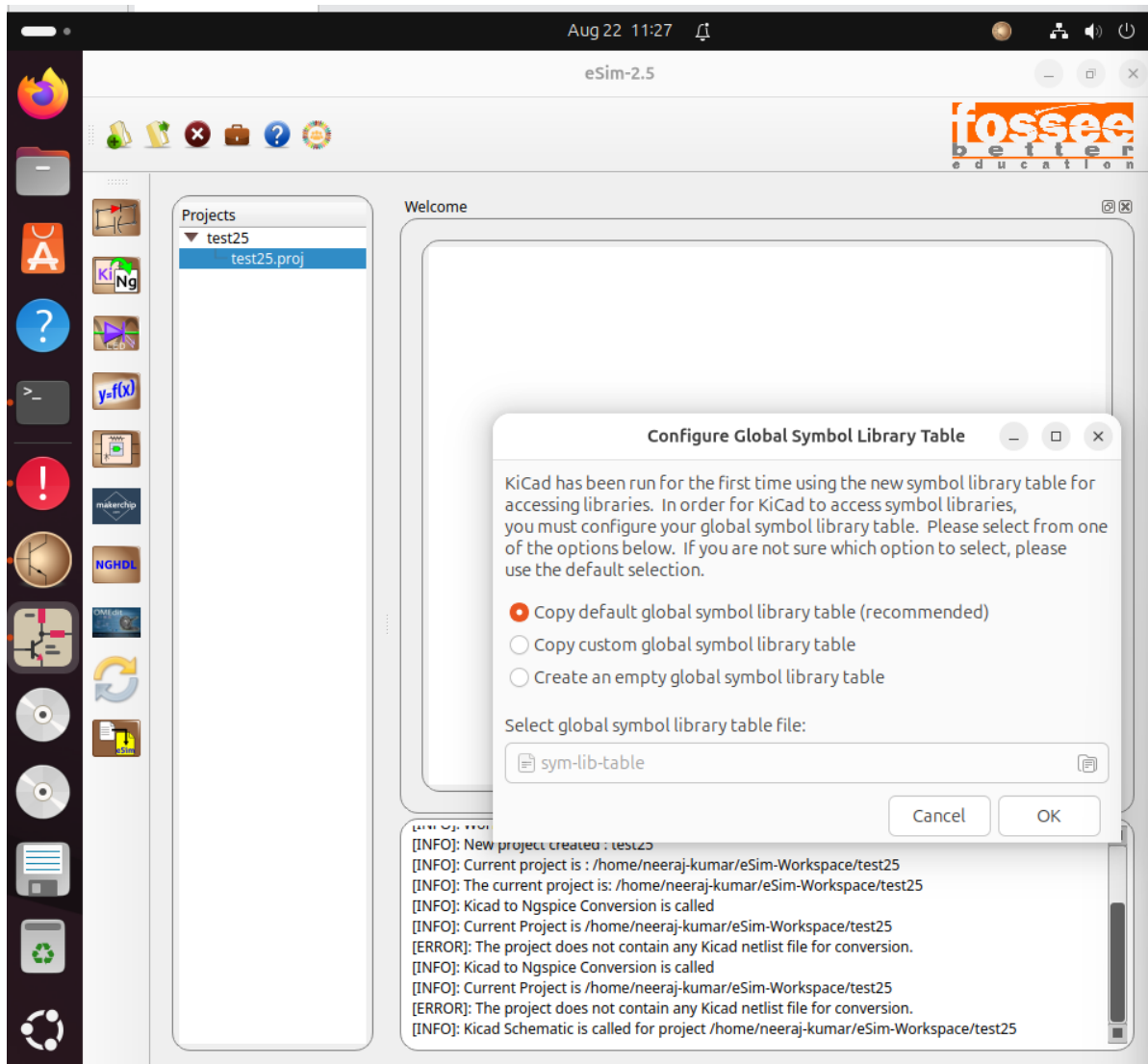


Figure 18. KiCad schematic editor opened for the eSim test project.

The project-to-KiCad path was reached successfully. The intermediate message that an empty project had no KiCad netlist was treated as expected behavior for a project with no schematic yet, not as a new installation defect.

```
$ git status
$ git log -1 --oneline
$ git push origin installers
```

Final repository state The installers branch was pushed to the fork `git@github.com:NeerajKumar011/eSim.git`. The working tree was left with only local untracked evidence files; those were not part of the source commit.

7. Quick Fix Guide

This section is the practical version intended for a user who encounters the same errors. It deliberately omits the investigation commands.

Error	Trigger	Minimal correction
-------	---------	--------------------

Error 1	Missing `fi`	Add `fi` after `bash "\$SCRIPT"\$ARGUMENT" in `install-eSim.sh`; run `bash -n install-eSim.sh`.
Error 2	`/: Is a directory`	Remove the seven misplaced configuration-write lines from the top-level `install-eSim.sh`; run `bash -n install-eSim.sh` and rerun the installer.
Error 3	Stops after KiCad is already installed	Change `exit 0` to `return 0` in `install-eSim-25.04.sh`; run `bash -n ...` and rerun.
Error 4	NGHDL rejects Ubuntu 25.04	Use `bash install-nghdl-scripts/install-nghdl-24.04.sh --install` in the 25.04 NGHDL function.
Error 5	GTK2 Canberra package unavailable	Remove `libcanberra-gtk-module` from the NGHDL dependency command.
Error 6	GHDL rejects LLVM 20.1.2	Add `check_version 20.0` and `check_version 20.1` to the GHDL configure version check.

8.Installation of Dependencies

Before the individual installation errors were encountered, the eSim installer checked and configured the required dependencies for the application and its simulation components.

The installation process included:

- Python virtual environment
- Make
- GNAT
- LLVM
- Clang
- Zlib development libraries
- X11/Athena widget libraries
- Build dependencies required by Verilator
- Build dependencies required by Ngspice
- PyQt5 and other Python components used by eSim

These dependencies provide the environment required for the eSim graphical interface, circuit simulation, and NGHDL/Verilator-based simulation components.

9.Final Result

After fixing the issues, eSim was installed successfully.

The application launched using:

The GUI opened correctly with all icons and functionalities available

10.Repository Link

GitHub Fork:

<https://github.com/NeerajKumar011/eSim>

<https://github.com/NeerajKumar011/eSim/tree/installers>

11.Conclusion

The installation of eSim 2.5 on Ubuntu 25.04 required a sequence of targeted compatibility corrections. The six documented errors were reproduced from the installation history and fixed one at a time.

- The top-level Bash syntax was corrected.
- Misplaced configuration writes were removed from the dispatcher.
- An already-installed KiCad instance no longer terminates the whole installer.
- Ubuntu 25.04 is routed through the bundled compatible NGHDL installer path.
- The obsolete GTK2 Canberra dependency is removed from the NGHDL dependency list.
- GHDL 4.1.0 was adapted to accept LLVM 20.0/20.1 and then passed a functional VHDL test.

The final environment successfully launched eSim, exposed KiCad 8.0.8, ran Verilator lint, performed a real Ngspice DC simulation, and produced a working GHDL 4.1.0 installation using LLVM 20.1.2.